

CIDO-FANet: Channel-Robust RF Fingerprinting Identification for 5G Devices

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Abstract—Radio-frequency fingerprinting identification (RFFI) can strengthen wireless security, but studies in 5G New Radio (NR) are scarce, and performance often degrades under channel variation. We propose a channel-robust RFFI framework for 5G NR. First, we introduce a Channel-Independent Differential Operator (CIDO) that derives channel-invariant device features from NR demodulation reference signals (DMRS), suppressing channel distortion while preserving device traits. Second, to accommodate NR's flexible resource mapping and exploit frequency-domain dependence, we design the Frequency-domain Autocorrelation Network (FANet), which handles variable-length frequency inputs and employs frequency-domain attention for improved discrimination. Finally, using data from five commercial 5G NR devices collected across diverse environments and times, our method attains consistently high accuracy and markedly better cross-channel generalization than the baselines. These results indicate a practical path toward reliable RFFI-based device identification in dynamic 5G NR settings.

Index Terms—5G NR, radio frequency fingerprinting, demodulation reference signal, channel-robust, deep learning

I. INTRODUCTION

5G and beyond cellular networks are expected to support massive connectivity, ultra-reliable low-latency services, and heterogeneous devices and deployments. At the same time, the broadcast nature of the wireless medium makes cellular systems vulnerable to diverse attacks that threaten privacy and security [1]. In current practice, many authentication and access-control mechanisms primarily rely on upper-layer identifiers and credentials. However, such upper-layer identifiers can be spoofed or replayed, enabling impersonation and other active attacks [2]. These risks motivate complementary identification mechanisms that are not limited to the upper layers and can provide device-centric evidence to strengthen trust decisions in real deployments.

Radio-frequency fingerprint identification (RFFI) is a physical-layer identification technique that exploits subtle, device-specific distortions caused by inevitable manufacturing

tolerances and hardware imperfections, which are difficult to replicate or counterfeit [3]. Conceptually, a transmitter leaves a characteristic “fingerprint” on emitted RF waveforms; the receiver learns to associate such fingerprints with device identities. Over the years, RFFI has been studied across multiple wireless technologies, including ZigBee [4], LoRa [5], Wi-Fi [6], [7], Bluetooth [8], and ADS-B [9]; in 4G-LTE, prior work has also considered access-related signals such as the physical random access channel (PRACH) [10]. Methodologically, the field has progressed from handcrafted RF descriptors to data-driven and end-to-end learning, where deep models learn discriminative fingerprints directly from signal representations [11]. Nonetheless, applying RFFI to 5G NR remains relatively under-explored. Compared with earlier systems, NR introduces flexible numerology and resource mapping, and it often operates in harsher propagation conditions, e.g., strong multipath, mobility, and dynamic scheduling, where channel effects can mask or distort device-specific signatures. These characteristics make *channel-robust* RFFI particularly important in 5G NR scenarios.

A key challenge is that many high-performing RFFI models implicitly rely on the i.i.d. assumption: training and test data share similar channel responses and noise statistics. In practice, transmitter signatures interact with time-varying propagation, and mismatches between training and deployment environments can cause severe distribution shifts and performance degradation [12]. Such channel sensitivity has been repeatedly observed; for example, notable accuracy drops have been reported when switching between LOS and NLOS conditions [7], and broader cross-environment generalization remains difficult [13]. Existing channel-robust strategies, including collecting data under diverse channel conditions [14], augmenting measurements with channel simulators [5], and adapting models with meta-learning and related techniques [15], are helpful but cannot be directly transferred to 5G NR without addressing NR-specific constraints. In particular, NR's protocol and waveform design can limit the availability and consistency of fingerprint-bearing observations; for example, dynamic and sparse reference signals, flexible scheduling and resource allocation, and varying transmission structures can reduce both the observability and stationarity of device-specific distortions, while practical deployments must respect standard procedures and operate under realistic receiver assumptions.

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Therefore, developing RFFI that is robust to channel variation and compatible with 5G NR protocol characteristics remains an open and practically important problem.

To address the above challenges, we proposed a channel-robust radio frequency fingerprint recognition scheme for 5G NR: designing channel-robust RF fingerprint features tailored to the characteristics of 5G NR, and developing a feature-matched network architecture to improve robustness under complex channel conditions; At the same time, system evaluation is completed across multiple channel conditions based on the data collected from a commercial 5G NR gNodeB(gNB). The main contributions are as follows:

- We propose a channel-suppressed RF fingerprint representation, termed CIDO, which preserves discriminative RFF cues without explicit channel equalization, offering low complexity and strong robustness.
- We propose FANet, which combines masked frequency-domain self-attention with length-invariant pooling to capture frequency-domain correlations and naturally support variable-length 5G NR inputs.
- We collect data using a commercial gNB and commercial 5G NR equipment across diverse wireless environments. Extensive experiments show that the proposed method enables reliable device identification under complex channels, thereby effectively supporting practical identity authentication.

The rest of this paper is organized as follows. The second part presents the system overview. The third part introduces the structure and preprocessing process of 5G NR frames. The fourth part defines the CIDO feature and introduces FANet, which leverages CIDO for channel-independent RFFI learning. The fifth part presents the experiment. Finally, the sixth part summarizes the full text.

II. SYSTEM OVERVIEW

This section provides an overview of the overall system architecture and the proposed algorithmic workflow. First, we describe the 5G NR system environment and data collection setup. Then, we outline the source-domain training and target-domain inference.

A. System Model

As shown in Fig. 1, our data collection is conducted in a commercial 5G NR system with multiple UEs communicating with a gNB in real wireless environments. The uplink transmissions experience diverse channel conditions before being received at the gNB. We focus on uplink access-related transmissions on the physical uplink shared channel (PUSCH). At the gNB side, we capture the received uplink baseband waveform and refer to it as the original signal (in Fig. 1), which serves as the raw input to the proposed CIDO-FANet RFFI Frame. By processing the original signal, the CIDO-FANet RFFI Frame identifies the transmitting device: for a legal UE, it allows network access and outputs the device identity, whereas for an illegal UE, it rejects access.

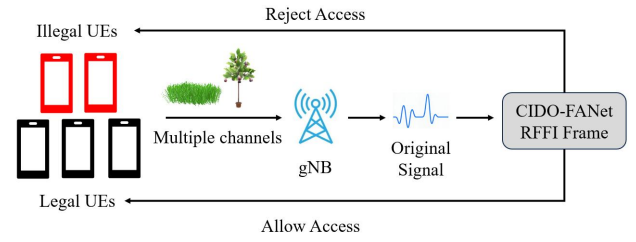


Fig. 1. 5G NR Data Collection Architecture.

B. CIDO-FANet RFFI Frame

As shown in Fig. 2, the proposed framework has two stages: source-domain training and target-domain inference. In source-domain training, the gNB captures tagged 5G NR uplink signals from multiple UEs and applies a unified pre-processing pipeline with data augmentation before feature construction. We then compute CIDO to obtain channel-suppressed RF fingerprint representations. FANet is trained as a feature extractor that maps CIDO representations to high-dimensional embeddings, which are stored in an RFF feature database. In target-domain inference, signals from unknown UEs undergo the same acquisition, preprocessing, and CIDO construction; FANet extracts embeddings that are matched against the database to output the UE identity. The next section details the 5G NR signal processing module.

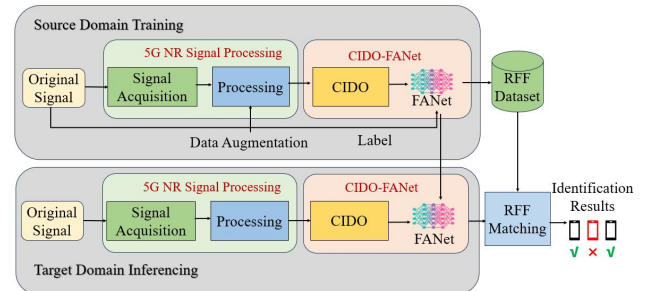


Fig. 2. CIDO-FANet RFFI Frame.

III. 5G NR SIGNAL PROCESSING

This section presents the essential 5G NR signal-processing pipeline with a specific goal: extracting demodulation reference signals (DMRS) from uplink PUSCH receptions. DMRS provides deterministic and repeatable observations that enable channel-robust RF fingerprint feature construction (CIDO) in the next section.

A. DMRS Position Selection in 5G NR for RFF Extraction

To reliably extract DMRS, we first summarize the NR time-frequency grid and clarify where DMRS resides. In 5G NR, a 10 ms frame comprises ten 1 ms subframes. The number of slots per subframe scales with the subcarrier spacing (SCS): 15/30/60/120/240 kHz correspond to 1/2/4/8/16 slots, respectively. With a normal cyclic prefix, each slot contains 14 OFDM symbols. In frequency, a resource block (RB) spans 12

subcarriers; therefore, one RB over one slot forms a 12×14 resource grid.

For uplink PUSCH, DMRS is a known sequence that can be generated at the receiver for channel estimation/compensation. Compared with earlier systems, NR offers flexible DMRS placement in both time and frequency. In time, DMRS symbols are located within the allocated PUSCH symbol set and can occupy one or multiple OFDM symbols, with additional positions configurable via standard parameters. In frequency, DMRS typically follows a comb-like occupancy whose density, ports, and CDM configuration determine its subcarrier pattern. In this work, the DMRS resource-element (RE) locations are identified strictly according to 3GPP TS 38.211 [16], and subsequent RF fingerprint extraction is aligned to these configured time–frequency positions.

B. Signal Acquisition and Preprocessing

At the gNB, we capture uplink baseband waveforms and apply a concise pipeline—triggering, synchronization, CFO compensation, OFDM demodulation, and DMRS extraction—to produce normalized DMRS observations that connect naturally to the CIDO formulation.

The continuous-time received baseband signal is modeled as

$$y(t) = h(\tau, t) * f(x(t)) + n(t), \quad (1)$$

where $x(t)$ is the ideal transmitted baseband signal, $f(\cdot)$ aggregates transmitter hardware impairments, $h(\tau, t)$ is the time-varying channel impulse response, $n(t)$ is additive noise, and $*$ denotes convolution over delay τ .

(1) Energy detection and triggering: we perform energy-based triggering on the streaming $y(t)$ to locate candidate uplink bursts for further processing.

(2) Timing synchronization: coarse timing is obtained via correlation/peak search to determine OFDM symbol boundaries (CP start), aligning symbols for demodulation.

(3) CFO estimation and compensation: CFO is estimated using CP-based autocorrelation and compensated by time-domain phase rotation to reduce inter-carrier interference.

(4) CP removal and FFT: after CP removal, an N -point FFT is applied to the l -th OFDM symbol, yielding

$$Y[k, l] = H[k, l] U[k, l] + N[k, l]. \quad (2)$$

where $k = 0, 1, \dots, N - 1$ indexes the subcarriers of the N -point FFT; $Y[k, l]$ is the demodulated RE on subcarrier k and OFDM symbol l ; $U[k, l]$ is the transmitted frequency-domain symbol (including transmitter impairments); $H[k, l]$ is the channel frequency response; and $N[k, l]$ denotes noise and residual demodulation errors. With a CP longer than the channel memory, the circular-convolution model holds.

(5) DMRS extraction and CSI Construction: according to scheduling and DMRS configuration, we select the DMRS RE set $\Omega_{\text{DMRS}} = \{(k, l)\}$ on the resource grid, whose positions are determined by the TS 38.211 PUSCH/DMRS mapping rules [16]. On these REs, since the ideal DMRS symbols

$S_{\text{ideal}}[k, l]$ are known at the receiver, we perform per-RE normalization to obtain a composite response

$$\text{CSI}(k, l) = \frac{Y[k, l]}{S_{\text{ideal}}[k, l]} = H[k, l] F_{\text{TX}}(k, l) + \tilde{N}[k, l]. \quad (3)$$

where $(k, l) \in \Omega_{\text{DMRS}}$; $S_{\text{ideal}}[k, l]$ denotes the known ideal DMRS symbol; $F_{\text{TX}}(k, l)$ aggregates transmitter-side fingerprint effects; and $\tilde{N}[k, l]$ is the normalized disturbance. Equation (3) makes explicit that the normalized DMRS observation still contains the channel term $H[k, l]$, motivating the channel-robust CIDO feature construction in the next section.

IV. CIDO-FANET CONSTRUCTION

This section constructs the channel-independent differential operator (CIDO). As shown in (3), the DMRS-normalized observation remains strongly colored by the frequency-selective channel, so directly using $\text{CSI}(k, l)$ may overfit environment-specific patterns and degrade under domain shift. CIDO is therefore designed to suppress channel coloration while preserving transmitter-dependent micro-dynamics.

A. Feature Construction of CIDO

In 5G NR, a single slot may contain multiple DMRS-bearing OFDM symbols. To exploit these additional observations, we generalize CIDO from a single quotient to a set of quotients formed by consecutive DMRS symbols, introducing a “symbol-pair” dimension. When the number of DMRS symbols within a slot is $M \geq 2$, this construction yields $M - 1$ quotients.

Let the DMRS-bearing symbol indices within the same slot be $\mathcal{L}_{\text{DMRS}} = \{l_1, l_2, \dots, l_M\}$ with $l_1 < l_2 < \dots < l_M$. For the p -th adjacent DMRS symbol pair (l_p, l_{p+1}) , the CIDO feature on a DMRS subcarrier k is defined as

$$\text{CIDO}_p(k) = \begin{cases} \frac{\text{CSI}(k, l_{p+1})}{\text{CSI}(k, l_p)}, & \text{CSI}(k, l_p) \neq 0, \\ c_{\text{max}}, & \text{CSI}(k, l_p) = 0, \end{cases} \quad (4)$$

where $p = 1, 2, \dots, M - 1$ and c_{max} is a sentinel value for the zero-denominator case. In implementation, we set c_{max} to the maximum 32767, so that subsequent stages can explicitly recognize these abnormal locations. This design avoids undefined quotients while preserving the information that the corresponding tone is unreliable, and it can be combined with reliability masking if desired.

Substituting (3) into (4) (excluding the zero-denominator special case) yields

$$\text{CIDO}_p(k) = \frac{H[k, l_{p+1}] F_{\text{TX}}(k, l_{p+1}) + \tilde{N}[k, l_{p+1}]}{H[k, l_p] F_{\text{TX}}(k, l_p) + \tilde{N}[k, l_p]}. \quad (5)$$

This expression shows that each quotient couples the channel evolution across the two symbols with the transmitter-dependent term.

When two DMRS symbols are close in time, the channel typically varies slowly over their separation. Under this short-time condition, the channel responses across the two symbols can be related by

$$H[k, l_{p+1}] \approx \gamma_p H[k, l_p], \quad (6)$$

where γ_p is approximately common across subcarriers and mainly reflects symbol-common gain/phase drift. Combining (6) with (5) indicates that the frequency-selective channel shape $H[k, l_p]$ largely cancels in the quotient. As a result, $\text{CIDO}_p(k)$ becomes less sensitive to multipath-induced spectral coloration while retaining transmitter-dependent variations.

The final CIDO feature vector is constructed by horizontally concatenating the quotient vectors in ascending pair order p , from (l_1, l_2) through (l_{M-1}, l_M) :

$$\mathbf{c} = [\mathbf{c}_1^\top, \mathbf{c}_2^\top, \dots, \mathbf{c}_{M-1}^\top]^\top, \quad \mathbf{c}_p = [\text{CIDO}_p(k)]_{k \in \mathcal{K}_{\text{eff}}}. \quad (7)$$

Here, $\mathcal{K}_{\text{eff}}$ denotes the set of retained DMRS tones after basic reliability screening, which may exclude tones with extremely small $|\text{CSI}(k, l_p)|$ to mitigate noise amplification and reduce the incidence of ill-conditioned quotients. The resulting concatenated vector preserves the subcarrier-wise structure within each quotient and incorporates multi-symbol temporal diversity via the symbol-pair dimension, providing richer inputs for FANet.

The effectiveness of this construction relies on the short-time relationship in (6). Under fast time variation (high Doppler), large DMRS spacing, or strong time selectivity, $H[k, l_{p+1}]/H[k, l_p]$ becomes strongly k -dependent, weakening cancellation and degrading feature stability and identification accuracy. Deep fades also increase the chance of zero/near-zero denominators and amplify disturbance terms, motivating reliability screening and sentinel marking. Overall, CIDO is best suited for low-to-moderate mobility with closely spaced DMRS symbols; under fast fading, using nearer DMRS pairs and aggregating multiple quotients can alleviate degradation.

B. FANet Construction

With the CIDO feature $\text{CIDO}(k)$ constructed in the previous subsection, we design a feature-aware network (FANet) to encode frequency-dependent transmitter-related features. Since $\text{CIDO}(k)$ is complex-valued, we decouple it into amplitude and phase sequences on the retained subcarriers:

$$x^{(a)}(k) = |\text{CIDO}(k)|, \quad k \in \mathcal{K}_{\text{eff}}, \quad (8)$$

$$x^{(p)}(k) = \arg \text{CIDO}(k), \quad k \in \mathcal{K}_{\text{eff}}. \quad (9)$$

Before specifying the network modules, we report frequency-domain ACF statistics computed along the subcarrier axis for the amplitude and phase components over 5 devices and 500 samples; markers show the mean ACF and the shaded envelope indicates mean $\pm$ std. NR supports flexible time-frequency resource mapping, and CIDO further applies retained-tone screening; consequently, the input length is variable. We denote by $K = |\mathcal{K}_{\text{eff}}|$ the number of retained subcarriers, and index them as a length- K sequence. Based on 5 devices and 500 samples, the amplitude sequence exhibits a high small-lag ACF that rapidly decays with increasing subcarrier lag, whereas the phase sequence shows near-zero ACF even at small lags. This indicates pronounced local

frequency dependence in amplitude but much weaker locality in phase, motivating asymmetric stream modeling together with variable-length accommodation.

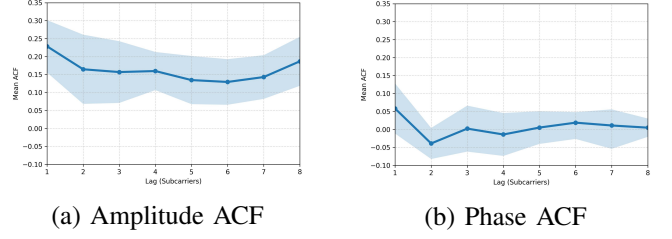


Fig. 3. Frequency-lag autocorrelation of CIDO components.

a) Attention mechanism design: FANet adopts ResNet-18 as the backbone and converts it into a 1D network for frequency-sequence processing. Specifically, all 3×3 2D convolutions in ResNet-18 are replaced by 3-tap 1D convolutions, and 2D pooling/downsampling is replaced by 1D pooling or strided 1D convolutions, while preserving the original residual stages and connections. The input is a multi-channel 1D sequence; thus the first convolution is configured with $\text{in_channels} = C_{\text{in}}$, where C_{in} is determined by the amplitude/phase streams.

Guided by the observed locality of amplitude, we introduce windowed local self-attention on the amplitude stream to emphasize informative neighboring subcarriers and suppress weakly related distant tones. For the phase stream, we avoid enforcing strong neighborhood mixing and instead employ a lightweight 1D stabilizer composed of pointwise (1×1) and shallow depthwise 1D convolutions with residual connections, improving stability without diluting phase variations.

Let $\mathbf{X}^{(a)} \in \mathbb{R}^{C_a \times K}$ denote amplitude features over K retained subcarriers. We form $\mathbf{Q}, \mathbf{K}_{\text{att}}, \mathbf{V}_{\text{att}} \in \mathbb{R}^{d \times K}$, with columns $\mathbf{q}_i, \mathbf{k}_i, \mathbf{v}_i \in \mathbb{R}^d$. Here, the position indices i and j refer to retained-subcarrier locations along the same length- K frequency sequence. Given the neighborhood $\mathcal{N}_r(i) = \{j : |j - i| \leq r\}$, the windowed self-attention output at position i is

$$\mathbf{o}_i = \sum_{j \in \mathcal{N}_r(i)} \alpha_{ij} \mathbf{v}_j, \quad \alpha_{ij} = \frac{\exp(\langle \mathbf{q}_i, \mathbf{k}_j \rangle / \sqrt{d})}{\sum_{j' \in \mathcal{N}_r(i)} \exp(\langle \mathbf{q}_i, \mathbf{k}_{j'} \rangle / \sqrt{d})}. \quad (10)$$

Windowing reduces the interaction cost from global $O(K^2)$ to $O(K(2r+1))$, consistent with the short-lag correlation structure of the amplitude stream.

b) Variable-length handling: Because NR resource mapping and the retained-tone set may vary, FANet must output a fixed-dimensional representation independent of K . Let the branch outputs be $\mathbf{Z}^{(a)} \in \mathbb{R}^{C_a \times K}$ and $\mathbf{Z}^{(p)} \in \mathbb{R}^{C_p \times K}$. We concatenate them into $\mathbf{F} = [\mathbf{Z}^{(a)}; \mathbf{Z}^{(p)}] \in \mathbb{R}^{C \times K}$, with $C = C_a + C_p$. With a mask $\mathbf{m} \in \{0, 1\}^K$ indicating valid subcarriers, we apply masked global average pooling:

$$g_c = \frac{\sum_{k=1}^K m_k F_{c,k}}{\sum_{k=1}^K m_k + \varepsilon}, \quad c = 1, \dots, C, \quad (11)$$

where $F_{c,k}$ is the (c,k) -th entry of $\mathbf{F}$ and $\varepsilon > 0$ prevents division by zero. The resulting embedding $\mathbf{g} = [g_1, \dots, g_C]^\top$ is invariant to K , accommodating different NR resource mappings.

V. EXPERIMENT

A. Experimental Setup

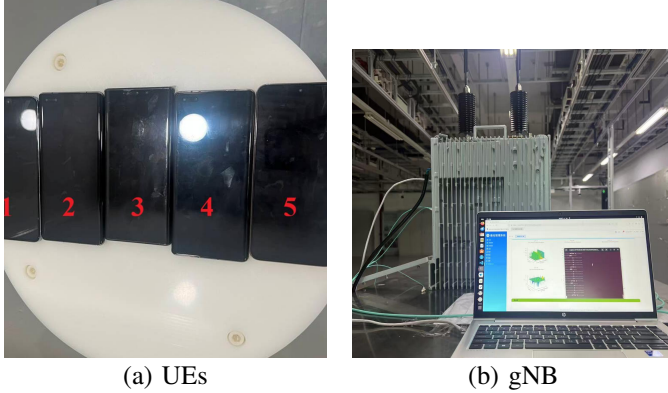


Fig. 4. Experimental equipment.

As shown in Fig. 4(a), we use five commercial available 5G NR devices as the target UEs; Devices 2 and 4 are from the same brand and model. All devices operate in compliance with 3GPP specifications. The gNB in Fig. 4(b) is an XTIBS1021E40 base station configured according to 3GPP Release 15, with a 15 kHz subcarrier spacing and a 20 MHz channel bandwidth.

Experiments are conducted in real-world wireless environments and organized into six domains. The indoor office environment $E1$ is an enclosed room ($8 \text{ m} \times 6 \text{ m}$), where the UE is placed at three locations with a fixed receiver: $E1_L1$ corresponds to a clear line-of-sight (LOS) link, while $E1_L2$ and $E1_L3$ are non-line-of-sight (NLOS) placements with increased blockage/multipath. The semi-open environment $E2$ is an industrial-like area ($20 \text{ m} \times 16 \text{ m}$); its two static domains $E2_L1$ and $E2_L2$ are also NLOS settings, and the mobile domain $E2_M$ captures UE motion under NLOS at approximately 10 km/h.

To evaluate cross-domain generalization, we use a leave-one-domain-out protocol. For each target domain, we train two models on the other two domains in the same scenario group (e.g., test $E1_L1$ and train on $E1_L2$ and $E1_L3$), and report the average of the two accuracies on the target.

The reported Test Accuracy is the classification accuracy. For a test set of N samples with ground-truth labels y_i and model output logits $\mathbf{o}_i$, it is defined as

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{I} \left(\arg \max_k (\mathbf{o}_i)_k = y_i \right), \quad (12)$$

where $\mathbb{I}(\cdot)$ is the indicator function.

B. Analysis of Experimental Results

Fig. 5 presents t-SNE visualizations of features extracted by CIDO-FANet at the six test domains under the evaluation

protocol described in Sec. V-A. Across all domains, samples from each device form clear, well-separated clusters. Cluster compactness is highest under the LOS domain ($E1_L1$), whereas the NLOS and semi-open domains ($E1_L2/E1_L3$, $E2_L1/E2_L2$) yield slightly looser yet still distinct groupings. Under the motion domain ($E2_M$), clusters exhibit a slight elongation. These visual patterns indicate that CIDO and FANet effectively suppress channel-induced variations while preserving device discriminability, demonstrating strong generalization across varying channel conditions.

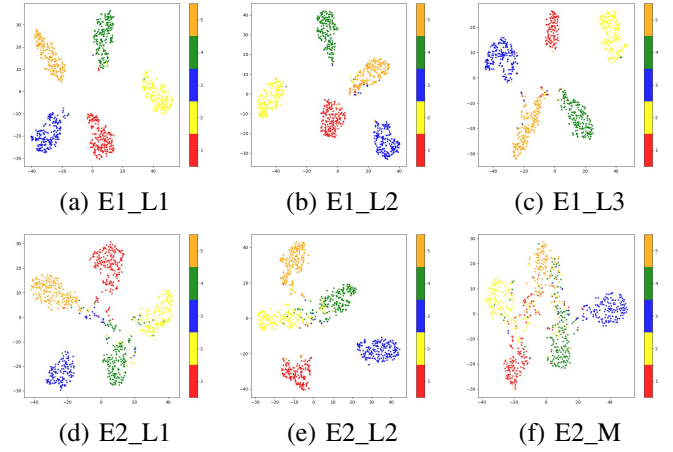


Fig. 5. Visualization of features extracted by CIDO-FANet.

Quantitatively, Fig. 6 summarizes the average cross-domain identification accuracy of four methods across all six test domains, including the proposed CIDO-FANet, the Channel Independent Spectrogram (ChInS) baseline [5], ZF one-tap channel equalization, and an I/Q baseline using the raw received I/Q samples. The results are obtained under the evaluation protocol described in Sec. V-A. CIDO-FANet attains the highest accuracies across all test domains—98.12%, 96.25%, and 97.27% at $E1_L1$, $E1_L2$, and $E1_L3$, and 94.70%, 91.71%, and 81.35% at $E2_L1$, $E2_L2$, and $E2_M$. On average, CIDO-FANet improves over ChInS by approximately 7.7 percentage points (maximum 10.43 at $E2_M$) and over ZF equalization by about 12.8 percentage points (maximum 25.19 at $E1_L2$). The margins over the raw I/Q baseline exceed 40 percentage points in all test domains, reaching as high as 45.51 percentage points in the $E2_M$ scenario. As channel conditions become more challenging and include motion ($E2_M$), accuracy naturally decreases due to stronger multipath, path loss, and time selectivity; nevertheless, CIDO-FANet degrades gracefully and maintains clear advantages. Overall, these quantitative results align with the visual evidence, corroborating the channel-robust cross-domain generalization capability of the proposed CIDO-FANet method.

To quantify FANet's contribution, we conduct an ablation study by removing FANet and using CIDO features alone. As shown in Table I, CIDO-FANet consistently outperforms the CIDO baseline across all six target domains, with an average gain of about 2.6 percentage points. The improvement is stable

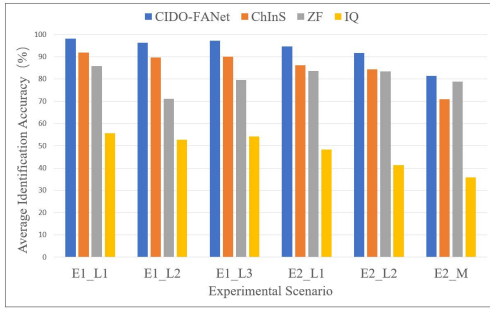


Fig. 6. Average identification accuracy with four methods.

in the office domains ($E1_L1$ – $E1_L3$; +2.16 to +2.84 points) and more pronounced in the semi-open domains (+2.39 at $E2_L1$ and +4.52 at $E2_L2$). Under the most challenging motion condition $E2_M$, FANet still yields a +2.83-point boost (81.35% vs. 78.52%), indicating improved robustness to time-varying channels. Overall, FANet’s dual-path design better exploits complementary amplitude and phase cues in CIDO, leading to more reliable identification under complex propagation.

TABLE I
CIDO-FANet vs CIDO Accuracy (%)

Domain Method	E1_L1	E1_L2	E1_L3	E2_L1	E2_L2	E2_M
CIDO-FANet	98.12	96.25	97.27	94.70	91.71	81.35
CIDO	95.43	94.09	94.46	92.31	87.19	78.52

VI. CONCLUSION

We presented CIDO-FANet, a channel-robust RFFI framework for 5G NR, addressing the scarcity of 5G RFFI studies and performance degradation under channel variation. It integrates two key components: the CIDO extracts channel-invariant features from NR DMRS, suppressing distortion while preserving device traits; the FANet adapts to NR’s flexible resource mapping and enhances discrimination via frequency-domain attention. Validated with data from five commercial 5G devices across diverse environments and times, CIDO-FANet achieves consistently high accuracy and superior cross-channel generalization, outperforming ChInS, ZF equalization, and the IQ baseline by 7.7, 12.8, and over 40 percentage points on average. These results fill the gap in practical 5G RFFI solutions, offering a viable path for reliable device identification in dynamic NR settings. Future work will expand NR configurations and explore real-time deployment.

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